Multiagent Systems
Agent Oriented Software Engineering

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Outline

1 Introduction

Introduction

- F. Zambonelli, A. Omicini. *Challenges and Research Directions in Agent-Oriented Software Engineering*. Journal of Autonomous Agents and Multi-Agent Systems, 9(3), pp. 253-283, 2004
- P. Giorgini, B. Henderson-Sellers. *Agent-Oriented Methodologies: An Introduction*. In Agent-Oriented Methodologies. December 2004 - January 2005. P. Giorgini and B. Henderson-Sellers (Eds.)

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Introduction

The features of agent-based systems are well suited to tackle the complexity of developing software in modern scenarios:

1. the autonomy of application components reflects the intrinsically decentralised nature of modern distributed systems and can be considered as the natural extension to the notions of modularity and encapsulation for systems that are owned by different stakeholders;

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2. the flexible way in which agents operate and interact (both with each other and with the environment) is suited to the dynamic and unpredictable scenarios where software is expected to operate;
3. the concept of agency provides for a unified view of AI results and achievements, by making agents and MASs act as sound and manageable repositories of intelligent behaviours.

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Introduction

- In the last few years, together with the increasing acceptance of agent-based computing as a novel software engineering paradigm, there has been a great deal of research related to the identification and definition of suitable models and techniques to support the development of complex software systems in terms of MASs.
- This research, can be roughly grouped under the term Agent-Oriented Software Engineering (AOSE).

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Section removed

The section “The Promise of AOSE for distributed systems engineering” has been removed in the version of the slides shortened in December 2023.

Current directions in AOSE

Mainstream research directions in AOSE:

1. Agent-Oriented Architectures
2. AOSE methodologies
3. Notation techniques
4. Agent-Oriented Programming Languages
5. Agent-based Modeling and Simulation
6. MAS infrastructures